Methods 8

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Apologies for the late submission! You can view my repo's commit history here if that helps to verify that this is my own work done independent from the solutions.

1.

a.

$$\text{Provided that :} \qquad \bar{Y}(A=0,B=0)=1, \qquad \bar{Y}(A=1,B=0)=5,$$

$$\bar{Y}(A=0,B=1)=4, \qquad \bar{Y}(A=1,B=1)=7$$

we can hold the following to be true.

$$\begin{aligned}\bar{Y}(A=0,B=0) &= \mathbb{E}(Y_i \mid A=0,B=0)\\ &= \mathbb{E}(\beta_0+\beta_1\cdot 0+\beta_2\cdot 0+\beta_3\cdot 0+\epsilon_i)\\ &= \beta_0+0+0+0+\mathbb{E}(\epsilon_i)\\ &= \beta_0+0=1\\ &\therefore \beta_0=1\end{aligned}$$

$$\begin{aligned}\bar{Y}(A=1,B=0) &= \mathbb{E}(Y_i \mid A=1,B=0)\\ &= \mathbb{E}(\beta_0+\beta_1\cdot 1+\beta_2\cdot 0+\beta_3\cdot 0+\epsilon_i)\\ &= \beta_0+\beta_1+0+0+\mathbb{E}(\epsilon_i)\\ &= 1+\beta_1+0=5\\ &\therefore \beta_1=4\end{aligned}$$

$$\begin{aligned}
\bar{Y}(A=0,B=1) &= \mathbb{E}(Y_i \mid A=0,B=1) \\
&= \mathbb{E}(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 1 + \beta_3 \cdot 0 + \epsilon_i) \\
&= \beta_0 + 0 + \beta_2 + 0 + \mathbb{E}(\epsilon_i) \\
&= 1 + \beta_2 + 0 = 4 \\
&\therefore \beta_2 = 3
\end{aligned}$$

$$\begin{aligned}
\bar{Y}(A=1,B=1) &= \mathbb{E}(Y_i \mid A=1,B=1) \\
&= \mathbb{E}(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot 1 + \beta_3 \cdot 1 + \epsilon_i) \\
&= \beta_0 + \beta_1 + \beta_2 + \beta_3 + \mathbb{E}(\epsilon_i) \\
&= 1 + 4 + 3 + \beta_3 = 7 \\
&\therefore \beta_3 = -1
\end{aligned}$$

Final Answer: $\boxed{\beta_0 = 1, \beta_1 = 4, \beta_2 = 3, \beta_3 = -1}$

b.

In order to get estimates for the four regression coefficients, we need estimates for the four averages above. We have two variables, A and B, which can take values 0 or 1 depending on whether or not the value observed was at a "high" or "low" level for each. β_0, the intercept or "baseline", necessitates that we have a treatment group where $A = 0, B = 0$. Then, by comparing to our baseline, we can get our treatment effects for A and B by changing the level one at a time: one group is $A = 1, B = 0$, and the other $A = 0, B = 1$. This lets us estimate β_1 and β_2. We also have to estimate the interaction effect β_3, so we need treatment group $A = 1, B = 1$. This constructs our full factorial design, where we have 2^2 treatment groups, one for each combination of treatment levels. Then, our observational units are randomly assigned to evenly split among the treatment groups.

In summary:

Group	Treatment	Number of units
Group 1	$A = 0, B = 0$	$n = 50$
Group 2	$A = 1, B = 0$	$n = 50$
Group 3	$A = 0, B = 1$	$n = 50$
Group 4	$A = 1, B = 1$	$n = 50$

c.

A full factorial design would be better:

We need to compare $A = 0,\ B = 0,\ C = 0$ and $A = 0,\ B = 0,\ C = 1$ to see C's treatment effect; i.e. how it changes the predicted result when all other variables are held constant. Then, to see how the effect changes for A and B, we need to compare $A = 0,\ B = 0,\ C = 1$ versus $A = 1,\ B = 0,\ C = 1$; likewise for B, we need to compare \$ A = 0, ~B = 0, ~C = 1\$ to $A = 0,\ B = 1,\ C = 1$. Assuming that our treatment groups from **(b)** had level $C = 0$, then this introduces 3 new groups. That gives us 7 treatment groups to estimate the interaction effects described.

A full factorial design would also include a three-way interaction effect, getting us $7 + 1 = 2^3$ groups, each group getting 25 units through random assignment. While a three-way interaction effect was not mentioned in the prompt, we also don't have reasonable certainty to assume that interaction effect to be zero. So, if we did a partial factorial design, we would be makiing the dubious assumption that there is no three-way interaction effect, and we would also be missing the interactions between C and A or B, or we would be missing treatment effects.